Supplementary Material S5

Screening Reconstruction and Corpus Update

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document describes the reconstruction of the identification and screening flow requested in the advisory review of 4 November 2026, and the resulting eligible corpus of 31 studies. Supplementary Materials S1 to S4 were updated to the same corpus in version V3.6. Section A describes the deduplication; Section B, the screening log and the explicit scope criterion; Section C, the decision on each of the 99 reports assessed in full text; Sections D and E, the extraction and MMAT appraisal of the 17 added studies; Section F, the retracted or withdrawn records; and Section G, the categories of implementation and validation evidence of the added studies.

Section A. Deduplication

All 1,920 exported records (Scopus 935, Web of Science 942, IEEE Xplore 38, ACM Digital Library 5) were processed with `pipeline/deduplicate_records.py`: exact match on the normalised DOI first, then on the normalised title and publication year, with precedence Scopus, Web of Science, IEEE Xplore and ACM Digital Library. The result is 524 duplicates, all matched by DOI and none by title and year, and 1,396 unique records, listed in `data/records_deduplicated.csv`.

The V3.5 flow reported 1,002 duplicates and 913 screened records. The reconstruction shows that the 913 contained 912 unique records plus one undetected duplicate (10.3390/jcm9072306), and that 479 unique records had been removed before screening without a documented reason. These 479 records were screened in V3.6. The five ACM records, originally excluded for lack of open access, were assessed by content because the ACM Digital Library has been fully open access since 1 January 2026 and the ACM search was run on 15 February 2026.

Section B. Screening log

Every unique record has one row in `data/screening_log.csv` with its stage, decision, criterion and reason. Title/abstract exclusions from the V3.5 screening ($n = 827$) keep their original decision; their reasons were not recorded at the time. The 479 omitted records were screened at title/abstract level in V3.6 (470 excluded, 9 assessed in full text). The 51 full-text exclusions of V3.5, whose reasons had not been recorded, were re-assessed in full text; the former criterion of fewer than 10 participants was removed, and sample size is now charted as a variable.

The scope criterion CI(1) was made explicit: an eligible study must use or evaluate an automated analysis of the face or gaze. The criterion was applied to every study that met all other criteria, including the V3.5 corpus. It excluded eight studies without any automated facial or gaze analysis: Reed et al. (2021) and Almeida et al. (2019) from the V3.5 corpus, and Chitti et al. (2025), Abd El-Sattar (2023), Crisci et al. (2025), Zhang et al. (2019), Cheng et al. (2016) and Palmer et al. (2023) from the re-screening. This is a deviation from the protocol and is reported as such in the manuscript.

Criterion CI(3) was audited against the SCImago edition of the year of first online publication of every included study and of every record moved to contextual evidence or excluded for CI(3); the result is recorded in `data/sjr_verification.csv`. All 31 eligible studies have a quartile from Q1 to Q3 in that edition; for the 2026 study the 2025 edition is used, because the 2026 edition has not been released. Five studies of the V3.5 corpus had no quartile in their year and were moved to contextual evidence, as Akın-Bülbül et al. (2024) had been: Rudovic et al. (2018) (Science Robotics, first quartile in 2019), Dapogny et al. (2019) (Frontiers in Computer Science,

first quartile in 2022), Rudovic et al. (2019) (ICMI 2019 proceedings), Sholikah et al. (2024) (Procedia Computer Science, a proceedings source) and Sariyanidi et al. (2025) (IEEE FG 2025, not indexed). The same audit corrected the reason of IEEE-0006: the 2025 edition of IEEE Access applies to it as to Cicceri et al. (2026), so it meets CI(3), and it is excluded for CE(3) because it uses only health-insurance administrative records.

Stage	*n*
Records identified	1,920
Duplicates removed	524
Records screened	1,396
Excluded at title/abstract	1,297
Reports assessed in full text	99
Reports excluded in full text	54
Contextual evidence documented separately	14
Studies included	31

Full-text exclusion criterion	*n*
CI(1)	16
CE(3)	10
CI(1) no automated facial or gaze analysis	8
Retracted or withdrawn	8
CI(3)	7
CE(2)	2
CI(4)	2
CE(1)	1
Total	54

Section C. Full-text decisions (99 reports)

Reports are ordered by decision (included, contextual evidence, retracted or withdrawn, excluded) and then by criterion and year. *Round* indicates whether the decision comes from the original V3.5 screening, from the re-assessment of the V3.5 full-text exclusions, or from the screening of the 479 records omitted before screening.

Record	**DOI**	**Title**	**Year**	**Decision**	**Criterion**	**Reason**	**Round**
WOS-0157	10.3390/s18113993	Computational Assessment of Facial Expression Production in ASD Children	2018	Included		Eligible corpus (V3.5): Leo et al. (2018)	V3.5
SCO-0895	10.1016/ j.artmed.2019.06.004	Labeling images with facial emotion and the potential for pediatric healthcare	2019	Included		D1: Kalantarian et al. 2019 (Artificial Intelligence in Medicine)	V3.6 re-assessment
SCO-0898	10.1001/ jamapediatrics.2019.0285	Effect of Wearable Digital Intervention for Improving Socialization in Children with Autism Spectrum Disorder: A Randomized Clinical Trial	2019	Included		D3: Voss et al. 2019	V3.6 re-assessment
SCO-0930	10.3389/fpsyt.2019.00402	The characteristics of intelligence profile and eye gaze in facial emotion recognition in mild and moderate preschoolers with autism spectrum disorder	2019	Included		Eligible corpus (V3.5): He et al. (2019)	V3.5
SCO-0657	10.1017/ s0954579420000449	Callous-unemotional traits in youth with autism spectrum disorder (ASD): Replication of prevalence estimates and associations with gaze patterns when viewing fearful faces	2021	Included		D3: Carter Leno et al. 2020	V3.6 re-assessment
SCO-0659	10.3390/mti5100057	Fostering emotion recognition in children with autism spectrum disorder	2021	Included		D2: Silva et al. 2021; former N<10 exclusion	V3.6 re-assessment

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
SCO-0681	10.3389/fpsyg.2021.678052	Emotional Expression in Children With ASD: A Pre-Study on a Two-Group Pre-Post-Test Design Comparing Robot-Based and Computer-Based Training	2021	Included		Eligible corpus (V3.5): Lecciso et al. (2021)	V3.5
SCO-0710	10.3390/jcm10081776	Is smiling the key? Machine learning analytics detect subtle patterns in micro-expressions of infants with asd	2021	Included		D4: Alvari et al. 2021	V3.6 re-assessment
SCO-0717	10.1002/aur.2391	Digital Behavioral Phenotyping Detects Atypical Pattern of Facial Expression in Toddlers with Autism	2021	Included		D4: Carpenter et al. 2020	V3.6 re-assessment
WOS-0611	10.3389/fpsyt.2020.568997	Emotional vs. Neutral Face Exploration and Habituation: An Eye-Tracking Study of Preschoolers With Autism Spectrum Disorders	2021	Included		D3: Bochet et al. 2021	V3.6, omitted 479
SCO-0550	10.1145/3526111	Toward Personalized Affect-Aware Socially Assistive Robot Tutors for Long-Term Interventions with Children with Autism	2022	Included		D2: Shi et al. 2022; former N<10 exclusion	V3.6 re-assessment
SCO-0584	10.2196/26760	Improved Digital Therapy for Developmental Pediatrics Using Domain-Specific Artificial Intelligence: Machine Learning Study	2022	Included		Eligible corpus (V3.5): Washington et al. (2022)	V3.5
IEEE-0014	10.1109/tnsre.2022.3192431	A Multimodal Approach for Identifying Autism Spectrum Disorders in Children	2022	Included		D4: Han et al. 2022	V3.6 re-assessment
SCO-0210	10.3390/children11111306	Autism Identification Based on the Intelligent Analysis of Facial Behaviors: An Approach Combining Coarse- and Fine-Grained Analysis	2024	Included		D4: Chen et al. 2024	V3.6 re-assessment
SCO-0235	10.1007/ s00521–024–09966–7	AutYOLO-ATT: an attention-based YOLOv8 algorithm for early autism diagnosis through facial expression recognition	2024	Included		Eligible corpus (V3.5): Hosney et al. (2024)	V3.5
WOS-0684	10.30773/pi.2023.0315	A Study on the Screening of Children at Risk for Developmental Disabilities Using Facial Landmarks Derived From a Mobile-Based Application	2024	Included		Eligible corpus (V3.5): Hwang et al. (2024)	V3.5
SCO-0012	10.3390/s25247485	Late Fusion Model for Emotion Recognition from Facial Expressions and Biosignals in a Dataset of Children with Autism Spectrum Disorder	2025	Included		D1: Kiejdo et al. 2025	V3.6 re-assessment
SCO-0021	10.1186/ s13229–025–00685-x	Comparing three algorithms of automated facial expression analysis in autistic children: different sensitivities but consistent proportions	2025	Included		D1: Manelis-Baram et al. 2025	V3.6 re-assessment
SCO-0022	10.1016/j.chbr.2025.100800	Assistive facial expression recognition for children with autism using re-enactment	2025	Included		Eligible corpus (V3.5): Johnson et al. (2025)	V3.5
SCO-0050	10.18280/isi.301007	Hybrid Deep Learning Framework for Autism Detection in Children Using Facial Emotion Recognition	2025	Included		Eligible corpus (V3.5): Shaik et al. (2025)	V3.5
SCO-0063	10.58346/jowua.2025.i3.026	A Facial Emotion Recognition System for Children with Autism, Based on The Integration of 3D-CNN and LSTM	2025	Included		Eligible corpus (V3.5): Aljbori et al. (2025)	V3.5

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
SCO-0064	10.3390/app15179555	Emotion Recognition in Autistic Children Through Facial Expressions Using Advanced Deep Learning Architectures	2025	Included		Eligible corpus (V3.5): Radočaj and Martinović (2025)	V3.5
SCO-0078	10.3390/ diagnostics15131601	Robust Autism Spectrum Disorder Screening Based on Facial Images (For Disability Diagnosis): A Domain-Adaptive Deep Ensemble Approach	2025	Included		Eligible corpus (V3.5): Alam et al. (2025)	V3.5
SCO-0087	10.1016/j.ijhcs.2025.103486	EmoLand: Utilizing narrative animations, multilevel games, and affective computing to foster emotional development in children with autism spectrum disorder	2025	Included		D3: Fan et al. 2025 (EmoLand)	V3.6 re-assessment
SCO-0113	10.3390/app15063083	Applying a Convolutional Vision Transformer for Emotion Recognition in Children with Autism: Fusion of Facial Expressions and Speech Features	2025	Included		D1: Wang et al. 2025	V3.6 re-assessment
SCO-0119	10.5815/ijigsp.2025.01.01	Disorder Facial Emotion Recognition with Grey Wolf Optimized Gefficient-Net for Autism Spectrum	2025	Included		Eligible corpus (V3.5): Vedantham et al. (2025)	V3.5
SCO-0137	10.1109/ access.2024.3403087	m_AutNet-A Framework for Personalized Multimodal Emotion Recognition in Autistic Children	2025	Included		D1: Kurian and Tripathi 2024	V3.6 re-assessment
SCO-0162	10.1177/ 15500594251362402	Using EEG and Eye Tracking to Evaluate an Emotion Recognition iPad App for Autistic Children	2025	Included		D3: Wall et al. 2025	V3.6 re-assessment
WOS-0254	10.1186/ s12888–025–07725-z	Digital screening of children with ASD: diagnostic accuracy of emotion recognition and visual preference tasks	2025	Included		D4: Pliska et al. 2025	V3.6 re-assessment
IEEE-0008	10.1109/tnsre.2025.3546519	Automated Autism Assessment With Multimodal Data and Ensemble Learning: A Scalable and Consistent Robot-Enhanced Therapy Framework	2025	Included		Eligible corpus (V3.5): Hassan et al. (2025)	V3.5
IEEE-0021	10.1109/ access.2026.3659276	AI4ASC: A Hierarchical Multi-Level AI-Driven Framework for Gesture Analysis and Monitoring in ASC Children	2026	Included		Eligible corpus (V3.5): Cicceri et al. (2026)	V3.5
SCO-0182	10.1038/ s41598–024–83229–2	A real-time environmental translator for emotion recognition in autism spectrum disorder	2024	Contextual evidence	Adult sample	C3 Levy et al. (2024): participants exclusively adult; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
SCO-0240	10.11591/ ijece.v14i4.pp4034–4045	Deep learning approaches for recognizing facial emotions on autistic patients	2024	Contextual evidence	Adult sample	C1 El Rhatassi et al. (2024): participants exclusively adult; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
SCO-0059	10.14569/ ijacsa.2025.0160959	Hybrid Real Time Facial Emotions Recognition on Autistic Individuals	2025	Contextual evidence	Adult sample	C2 El Rhatassi et al. (2025): participants exclusively adult; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
SCO-0648	10.3390/brainsci11111531	Differences in performance of asd and adhd subjects facing cognitive loads in an innovative reasoning experiment	2021	Contextual evidence	CE(3)	C5 Papaioannou et al. (2021): exclusively non-facial data; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
SCO-0566	10.3390/s22145116	Understanding Emotions in Children with Developmental Disabilities during Robot Therapy Using EDA	2022	Contextual evidence	CE(3)	C4 Nagae and Lee (2022): exclusively non-facial data; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
WOS-0425	10.3390/brainsci13050779	Abnormal Brain Structure Is Associated with Social and Communication Deficits in Children with Autism Spectrum Disorder: A Voxel-Based Morphometry Analysis	2023	Contextual evidence	CE(3)	C6 Xu and Ju (2023): exclusively non-facial data; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
WOS-0435	10.1038/ s41598–023–36701–4	Automatic speaker diarization for natural conversation analysis in autism clinical trials	2023	Contextual evidence	CE(3)	C7 O'Sullivan et al. (2023): exclusively non-facial data; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
WOS-0292	10.1038/ s41598–025–09862–7	MDNCT: a multi-domain neurocognitive transformer architecture approach for early prediction of autism spectrum disorders	2025	Contextual evidence	CE(3)	C8 Alsubai (2025): exclusively non-facial data; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
WOS-0437	10.1126/ scirobotics.aao6760	Personalized machine learning for robot perception of affect and engagement in autism therapy	2018	Contextual evidence	CI(3)	C10 Rudovic et al. (2018): Science Robotics has no SJR quartile in 2018 (first quartile in 2019); documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	SJR audit V3.6 (2026–11–10)
SCO-0879	10.1145/3340555.3353742	Multi-modal active learning from human data: A deep reinforcement learning approach	2019	Contextual evidence	CI(3)	C12 Rudovic et al. (2019): ICMI 2019 proceedings without SJR quartile; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	SJR audit V3.6 (2026–11–10)

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
SCO-0880	10.3389/fcomp.2019.00005	On Automatically Assessing Children's Facial Expressions Quality: A Study, Database, and Protocol	2019	Contextual evidence	CI(3)	C11 Dapogny et al. (2019): Frontiers in Computer Science has no SJR quartile in 2019 (first quartile in 2022); documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	SJR audit V3.6 (2026–11–10)
SCO-0325	10.1016/j.procs.2024.02.169	Real-time Facial Expression Recognition to Enhance Emotional Intelligence in Autism	2024	Contextual evidence	CI(3)	C13 Sholikah et al. (2024): Procedia Computer Science is a proceedings source without SJR quartile; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	SJR audit V3.6 (2026–11–10)
WOS-0236	10.31828/ turkpsikoloji.1515430	Otizm Belirtilerinin Erken Tespitinde Duygu Durumlarına Yöneltilen Görsel Dikkatin Makine Öğrenmesi Aracılığıyla Değerlendirilmesi	2024	Contextual evidence	CI(3)	C9 Akın-Bülbül et al. (2024): journal without SJR quartile in the year of publication; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	V3.5
SCO-0160	10.1109/ fg61629.2025.11099288	Beyond FACS: Data-driven Facial Expression Dictionaries, with Application to Predicting Autism	2025	Contextual evidence	CI(3)	C14 Sariyanidi et al. (2025): IEEE FG 2025 proceedings not indexed in SJR; documented as contextual evidence (Supplementary Materials S2 and S4, Section B)	SJR audit V3.6 (2026–11–10)
WOS-0252	10.1155/2022/2247675	RETRACTED: An AI-Enabled Internet of Things Based Autism Care System for Improving Cognitive Ability of Children with Autism Spectrum Disorders (Retracted Article)	2022	Retracted	Retraction	Retracted 14 Sep 2023	V3.6, omitted 479
SCO-0190	10.1038/ s41598–024–77549–6	Ambiguous facial expression detection for Autism Screening using enhanced YOLOv7-tiny model	2024	Retracted	Retraction	Retracted article	V3.5
SCO-0244	10.1049/htl2.12073	Autism spectrum disorder detection using facial images: A performance comparison of pretrained convolutional neural networks	2024	Retracted	Retraction	Retracted article	V3.5
SCO-0271	10.1007/ s00500–023–09477-y	Real-time facial emotion recognition model based on kernel autoencoder and convolutional neural network for autism children	2024	Retracted	Retraction	Retracted 16 Nov 2025	V3.6 re-assessment
SCO-0013	10.1371/ journal.pone.0338701	Can artificial intelligence and face recognition using deep learning detect emotions in children with autism?	2025	Retracted	Retraction	Retracted article	V3.5
SCO-0026	10.1016/ j.rineng.2025.106989	ERAnet: Emotion Recognition based Assistive learning network for autistic children	2025	Retracted	Retraction	Article removed at the request of the Executive Editors	V3.6 re-assessment
SCO-0036	10.1038/ s41598–025–97708–7	A multi-filter deep transfer learning framework for image-based autism spectrum disorder detection	2025	Retracted	Retraction	Retracted 8 Dec 2025 (undocumented image provenance)	V3.6 re-assessment

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
SCO-0037	10.1038/s41598–025–96014–6	Leveraging artificial intelligence for diagnosis of children autism through facial expressions	2025	Retracted	Retraction	Retracted article	V3.5
WOS-0428	10.1016/j.pnpbp.2018.05.021	Emotions in action: The relationship between motor function and social cognition across multiple clinical populations	2018	Excluded	CE(1)	Narrative review	V3.6 re-assessment
SCO-0923	10.30534/ijatcse/2019/6381.62019	A deep learning approach in robot-assisted behavioral therapy for autistic children	2019	Excluded	CE(2)	No own quantitative metric	V3.6 re-assessment
SCO-0075	10.3390/computers14070292	A New AI Framework to Support Social-Emotional Skills and Emotion Awareness in Children with Autism Spectrum Disorder	2025	Excluded	CE(2)	No quantitative evaluation	V3.6 re-assessment
WOS-0733	10.1038/s41598–018–32454–7	An emotional modulation model as signature for the identification of children developmental disorders	2018	Excluded	CE(3)	Speech only	V3.6, omitted 479
SCO-0840	10.1109/access.2020.3012532	Estimating Autism Severity in Young Children from Speech Signals Using a Deep Neural Network	2020	Excluded	CE(3)	Speech only	V3.6 re-assessment
SCO-0620	10.1155/2022/4576211	Practice of Music Therapy for Autistic Children Based on Music Data Mining	2022	Excluded	CE(3)	Music emotion features only	V3.6 re-assessment
WOS-0415	10.3390/jintelligence10040112	Cognitive and Developmental Functions in Autistic and Non-Autistic Children and Adolescents: Evidence from the Intelligence and Development Scales-2	2022	Excluded	CE(3)	Cognitive test battery only	V3.6 re-assessment
IEEE-0012	10.1109/tnsre.2024.3417210	Characterizing Autism Spectrum Disorder Through Fusion of Local Cortical Activation and Global Functional Connectivity Using Game-Based Stimuli and a Mobile EEG System	2024	Excluded	CE(3)	EEG only	V3.6 re-assessment
IEEE-0018	10.13052/jicts2245–800x.1231	Enhancing Communication for People with Autism Through the Design of a Context-Aware Mobile Application for PECS	2024	Excluded	CE(3)	PECS communication app design; no facial or gaze analysis	V3.6 re-assessment
SCO-0172	10.1016/j.procs.2025.04.243	Emotion Recognition and Intervention Technology for Autistic Children Based on the Fusion of Neural Networks and Biological Signals	2025	Excluded	CE(3)	Heart rate, GSR and EEG only	V3.6 re-assessment
WOS-0620	10.1002/aur.70095	Characterizing the Relationship Among the Social Competence Elements in Autistic Adolescents	2025	Excluded	CE(3)	EEG/ERP and behaviour ratings only	V3.6 re-assessment
WOS-0022	10.3389/fpsyt.2025.1703302	Cortical hemodynamic responses and deep learning models of emotional face processing in preschool children with autism spectrum disorder: a fNIRS study	2026	Excluded	CE(3)	fNIRS only; faces are stimuli, not analysed	V3.6 re-assessment
IEEE-0006	10.1109/access.2026.3650856	A Data-Driven Approach for Autism Spectrum Disorder Screening	2026	Excluded	CE(3)	Health-insurance administrative records only (consultation and procedure requests); no facial or gaze input	SJR audit V3.6 (2026–11–10)
WOS-0068	10.1186/s13229–018–0187–7	Facial expression recognition as a candidate marker for autism spectrum disorder: how frequent and severe are deficits?	2018	Excluded	CI(1)	Predominantly adult sample (mean age 30; 2 of 46 under 18)	V3.6, omitted 479

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
WOS-0092	10.1080/20473869.2016.1206729	Investigating mobile emotional learning for children with autistic spectrum disorders	2018	Excluded	CI(1)	Cheng et al. 2016: no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
WOS-0400	10.2196/mental.9631	Improvement of Attention-Deficit/Hyperactivity Disorder Symptoms in School-Aged Children, Adolescents, and Young Adults With Autism via a Digital Smartglasses-Based Socioemotional Coaching Aid: Short-Term, Uncontrolled Pilot Study	2018	Excluded	CI(1)	Outcome is ADHD symptoms (outside D1-D4); former N<10 exclusion	V3.6 re-assessment
WOS-0744	10.1109/access.2018.2878144	Mean of Correlation Methodfor Optimization of Affective States Detection in Children	2018	Excluded	CI(1)	Healthy children only	V3.6, omitted 479
SCO-0878	10.1080/20473869.2019.1656384	Using technology-based learning tool to train facial expression recognition and emotion understanding skills of Chinese pre-schoolers with autism spectrum disorder	2019	Excluded	CI(1)	Zhang et al. 2019: no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
SCO-0894	10.35940/ijrte.b1095.0782s319	Recognition of emotion from facial expression for autism disorder	2019	Excluded	CI(1)	Trained and tested on CK+ only; no data from children with ASD	V3.6 re-assessment
SCO-0932	10.1155/2019/4384896	ALTRIRAS: A Computer Game for Training Children with Autism Spectrum Disorder in the Recognition of Basic Emotions	2019	Excluded	CI(1)	Almeida et al. 2019: no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
WOS-0416	10.1002/aur.2084	Psychometric Evaluation of Social Cognitive Measures for Adults with Autism	2019	Excluded	CI(1)	Adults only	V3.6 re-assessment
SCO-0841	10.1515/comp-2020–0006	AutisMitr: Emotion Recognition Assistive Tool for Autistic Children	2020	Excluded	CI(1)	Tested with typically developing children only	V3.6 re-assessment
IEEE-0023	10.1109/access.2020.3018738	Toward Unbiased Facial Expression Recognition in the Wild via Cross-Dataset Adaptation	2020	Excluded	CI(1)	General in-the-wild FER; no ASD sample	V3.6 re-assessment
SCO-0655	10.1111/jcpp.13395	Examining the bidirectional association between emotion recognition and social autistic traits using observational and genetic analyses	2021	Excluded	CI(1)	Reed et al. 2021: no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
ACM-0001	10.1145/3501712.3538831	Real-time Feedback based on Emotion Recognition for Improving Children's Metacognitive Monitoring Skill	2022	Excluded	CI(1)	Extended abstract of a PhD plan; no ASD sample	Re-screening V3.6
SCO-0454	10.3390/drones7020081	Deep Learning-Based Cost-Effective and Responsive Robot for Autism Treatment	2023	Excluded	CI(1)	No ASD sample reported (appearance survey of 60 people aged 15–62)	V3.6 re-assessment
SCO-0472	10.1109/tnsre.2023.3333049	Design Method of a Smart Rehabilitation Product Service System Based on Virtual Scenarios: A Case Study	2023	Excluded	CI(1)	Rehabilitation service system design; not ASD	V3.6 re-assessment
SCO-0495	10.14569/ijacsa.2023.0140414	EMOCASH: An Intelligent Virtual-Agent Based Multiplayer Online Serious Game for Promoting Money and Emotion Recognition Skills in Egyptian Children with Autism	2023	Excluded	CI(1)	Abd El-Sattar 2023 (EMOCASH): no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
WOS-0418	10.3389/fnins.2023.1295608	A neurophysiological signature of dynamic emotion recognition associated with social communication skills and cortical gamma-aminobutyric acid levels in children	2023	Excluded	CI(1)	Typically developing children only	V3.6 re-assessment

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
IEEE-0028	10.1109/access.2023.3294099	CNN Learning Strategy for Recognizing Facial Expressions	2023	Excluded	CI(1)	General FER (Tufts, RWTH, RAF, FER2013); no ASD sample	V3.6 re-assessment
ACM-0002	10.1145/3630970.3631080	Hybrid Models for Facial Emotion Recognition in Children	2023	Excluded	CI(1)	Typically developing children only	Re-screening V3.6
SCO-0256	10.1002/aur.3140	Smartphone-based gaze estimation for in-home autism research	2024	Excluded	CI(1)	Adults only; gaze estimation	V3.6, omitted 479
SCO-0281	10.1007/s10803–022–05831–5	Effect of Emotional Valence on Emotion Recognition in Adolescents with Autism Spectrum Disorder	2024	Excluded	CI(1)	Palmer et al. 2023: no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
SCO-0111	10.1016/j.chbr.2024.100549	MiEmo: A multi-modal platform on emotion recognition for children with autism spectrum condition	2025	Excluded	CI(1)	Chitti et al. 2025 (MiEmo): no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
SCO-0142	10.1007/s10803–025–07120–3	Facial Emotion Recognition in Neurodevelopmental Disorders: A Comparative Study in Children and Adolescents With and Without Autism, ADHD and Specific Learning Disorders	2025	Excluded	CI(1)	Crisci et al. 2025: no automated facial or gaze analysis (scope criterion made explicit in V3.6)	Scope criterion applied in V3.6 to all included studies
WOS-0584	10.1038/s41598–025–97701–0	Positive emotional valence in spontaneous facial expressions of autistic adolescents	2025	Excluded	CI(1)	Human-coded facial expressivity; no emotion recognition and no computational analysis	V3.6, omitted 479
ACM-0004	10.1145/3744257.3744274	Color-Coded Emotional Learning Serious Games for Children with Autism: Foregrounding Facial Expression Recognition & Accessibility	2025	Excluded	CI(1)	Opinion survey; no participants with ASD	Re-screening V3.6
SCO-0909	10.1007/s41666–018–0034–9	Guess What?: Towards Understanding Autism from Structured Video Using Facial Affect	2019	Excluded	CI(3)	Journal of Healthcare Informatics Research has no SJR quartile in 2019; former N<10 exclusion	V3.6 re-assessment
WOS-0615	10.1109/msmc.2021.3086989	AI-Augmented Behavior Analysis for Children With Developmental Disabilities: Building Toward Precision Treatment	2021	Excluded	CI(3)	Trade journal without quartile	V3.6 re-assessment
SCO-0534	10.1145/3575879.3575997	ToMtool: An Interactive Multimedia Application to Support Training of Emotion Recognition and Theory of Mind Skills to Children with Autism Spectrum Disorder	2022	Excluded	CI(3)	ACM ICPS proceedings without quartile	V3.6 re-assessment
SCO-0467	10.1109/aciiw59127.2023.10388145	Development and Validation of an iPad-based Serious Game for Emotion Recognition and Attention Tracking towards Early Identification of Autism	2023	Excluded	CI(3)	Workshop proceedings without quartile	V3.6 re-assessment
SCO-0489	10.5220/0011748800003470	Using Deep Learning and Native Mobile App to Assist Autistic Students' Educational Experience	2023	Excluded	CI(3)	Conference proceedings without quartile	V3.6 re-assessment
ACM-0003	10.1145/3613905.3651045	Early Autism Screening in Children Using Facial Recognition	2024	Excluded	CI(3)	CHI Extended Abstracts: proceedings source without SJR quartile	Re-screening V3.6
ACM-0005	10.1145/3660853.3660888	Deep Learning-Based Feature Extraction from Children's Facial Images for Autism Spectrum Disorder Detection	2024	Excluded	CI(3)	AICCONF 2024 not indexed in SJR	Re-screening V3.6

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Record	DOI	Title	Year	Decision	Criterion	Reason	Round
SCO-0866	10.1109/tg.2018.2864640	The ASC-inclusion perceptual serious gaming platform for autistic children	2019	Excluded	CI(4)	No open-access full text	V3.6 re-assessment
WOS-0119	10.1016/ j.displa.2025.103049	Autism screening for children based on appearance features across multiple paradigms	2025	Excluded	CI(4)	No open-access full text	V3.6, omitted 479

Section D. Studies added to the eligible corpus (17 studies): extraction

First-reviewer extraction from the full texts. The fields left pending in the first pass (Han et al., Carpenter et al., Kalantarian et al., Kurian and Tripathi, Pliska et al. and Fan et al.) were verified against the full texts on 9 November 2026. The advisory review requires confirmation by a second reviewer. Shi et al. (2022) was coded from its arXiv version; the classifier details of Voss et al. (2019) are in supplementary material that could not be accessed.

#	Study	Dom.	Country	Participants	Data and modalities	Method	Dataset	Validation	Main results
29	Kiejdo et al. (2025)	D1	Poland (data from Poland, Turkey, North Macedonia)	99 robot-session recordings of children with ASD aged 0–12; N children not reported	FaceReader outputs + EDA, skin temperature, heart rate (4)	Per-modality BiLSTM, decision-level late fusion	EMBOA (private)	Subject-disjoint 60/20/20 split	ACC 0.68, micro-F1 0.68, macro recall 0.64; likelihood scheme ACC 0.78
30	Wang et al. (2025)	D1	China	33 children with ASD (DSM-5), 4–5 years	Face video frames + speech (2)	CvT + MS-CAM attention fusion	Own (private)	Subject-wise 7:2:1 split	ACC face 79.12%, speech 83.47%, fusion 90.73%
31	Kurian and Tripathi (2024)	D1	India	SSBD (75 videos) plus additional YouTube videos	Face + audio (2); 4 classes (happy, sad, neutral, meltdown)	m_AutNet with WGAN domain adaptation	SSBD + YouTube (public web; diagnosis and consent not verifiable)	Hold-out test set of 80 samples; separation of children not reported	Test ACC 88.25%
32	Manelis-Baram et al. (2025)	D1	Israel	100 (72 ASD with ADOS-2 and DSM-5, 28 controls), 2–7 years	ADOS-2 video (1)	iMotions, FaceReader, Py-Feat (off-the-shelf)	NADI recordings (on request)	No training; cross-algorithm agreement	Agreement r 0.37–0.75; no ASD-control difference
33	Chen et al. (2024)	D4	China	81 (41 ASD, DSM-5 by two pediatricians; 40 age-matched TD), 4–6 years	Facial video (1)	Expression/head-pose descriptors + LSTM + SENet	Own (on request)	3-fold CV by participant	ACC 88.74%
34	Shi et al. (2022)	D2	USA	4 children with ASD, 19 in-home sessions, mean 4.6 years	Face/body video, audio, game logs (3+)	XGBoost with supervised domain adaptation	Own (access not reported)	Session-wise chronological split per child	AUROC arousal 0.92, valence 0.86; positive-class F1 0.90/0.89
35	Silva et al. (2021)	D2	Portugal	6 children with ASD, 8–9 years (classifier trained on 43 non-ASD participants)	RealSense 3D face (1)	SVM (RBF)	Own non-ASD training data (private)	10-fold CV (partition unit not reported)	ACC offline 93.63%; real time 88.3% (adults)

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#	Study	Dom.	Country	Participants	Data and modalities	Method	Dataset	Validation	Main results
36	Alvari et al. (2021)	D4	Italy	33 infants (18 ASD with ADOS-2, 15 TD), 6–12 months	Home video (1)	OpenFace action units; statistics only	Own (not shareable)	Not applicable	MANOVA F(6,22)=14.51, p<0.0001
37	Carpenter et al. (2020)	D4	USA	104 toddlers (22 ASD)	Tablet camera (1)	Computer vision features; LASSO-selected logistic regression per movie	Own clinical (private)	Leave-one-out cross-validation	AUC 0.62–0.73 per movie
38	Pliska et al. (2025)	D4	Germany	48 boys (24 ASD, 24 matched TD), 6–11 years	Tablet gaze + emotion task responses (2)	Decision trees (Gini) and mixed models	Own; CEFE stimuli	No cross-validation or hold-out reported	ACC 81.25%, sensitivity 91.67%, specificity 70.83%
39	Voss et al. (2019)	D3	USA	71 children with ASD (40 intervention, 31 control), 6–12 years	Real-time camera on Google Glass (1)	Embedded FER classifier (about 72% accuracy, eMethods)	Trial data	Randomized clinical trial	VABS-II socialization +4.58, p=0.005
40	Fan et al. (2025)	D3	China	12 children with ASD, 4–7 years	Real-time camera (1); 7 training classes, 4 evaluated	CNN trained on CAFE + FER-2013	CAFE + FER-2013 (public)	Pre/post; 34,221/8,141 train/test images	AI feedback accuracy 97.6% (specialist review); pre/post d 0.94–1.26
41	Wall et al. (2025)	D3	Australia	35 (15 ASD, 20 NT), 6–12 years	Eye-tracking of CAFE faces + ERP (2)	Statistics only	Own	Pre/post intervention	Whole-face fixation Z=-3.98, p<0.001
45	Carter Leno et al. (2020)	D3	UK	189 adolescents with ASD; 46 with eye-tracking	Eye-tracking of fearful faces	Statistics only	Cohort data	Cross-sectional	CU traits associated with fewer eye fixations on fearful faces
48	Kalantarian et al. (2019)	D1	USA	8 boys with ASD, mean 8.5 years, one session (Guess What? game)	Game video (1); 7 categories	Ensemble of commercial emotion classifiers for semi-automatic frame labelling	Own clinical	Not applicable (labelling against manual raters)	Frames labelled correctly: disgust 94%, neutral 81%, surprise 92%, scared 56%
49	Han et al. (2022)	D4	China	90 (40 ASD, 50 TD), 3–6 years	EEG + eye-tracking (2)	Multimodal stacked denoising autoencoders (MMSDAE)	Own	10-fold subject-independent stratified cross-validation	ACC fusion 95.56% (sensitivity 92.5%, specificity 98%); eye-tracking 86.67%, EEG 81.11%
51	Bochet et al. (2021)	D3	Switzerland	46 (24 ASD with ADOS, 22 TD), 1.8–5.8 years	Eye-tracking of emotional vs neutral faces (Tobii TX300)	Statistics only	Radboud Faces stimuli	Group comparison	Fewer fixations on emotional faces in ASD (p=0.004)

Section E. MMAT 2018 appraisal of the added studies

Categories: 2, randomised controlled trial; 3, quantitative non-randomised; 4, quantitative descriptive. The five judgement columns follow the aggregated domains used in Supplementary Material S2 and in the MMAT figure of the manuscript: **Rep.** = representativeness; **Meas.** = measurements; **Int.** = intervention as intended and sampling strategy; **Conf.** = confounding; **Out.** = outcome completeness. For randomised trials, randomisation is reported under confounding and adherence under

intervention as intended. Judgements are coded as **Y** = Yes, **N** = No, **ND** = Not Determinable.

#	Study	Cat.	Rep.	Meas.	Int.	Conf.	Out.	Coding notes
29	Kiejdo et al. (2025)	3	ND	Y	Y	Y	Y	EMBOA robot sessions; N children not reported; late fusion of per-modality BiLSTM
30	Wang et al. (2025)	3	Y	Y	Y	Y	Y	33 children with ASD aged 4–5, rehabilitation classroom; CvT face + speech with MS-CAM fusion; subject-wise 7:2:1 split; ACC 90.73% (95% bootstrap CI 89.86–91.60) with per-class precision and recall
31	Kurian and Tripathi (2024)	3	N	ND	ND	ND	ND	SSBD (75 children) plus YouTube videos, diagnosis and consent not verified; 4 classes (happy, sad, neutral, meltdown); hold-out test set of 80 samples, subject separation not reported; test ACC 88.25%
32	Manelis-Baram et al. (2025)	3	ND	Y	Y	Y	ND	Off-the-shelf FER tools (iMotions, FaceReader, Py-Feat) on ADOS-2 videos; no trained model
33	Chen et al. (2024)	3	Y	Y	Y	Y	ND	LSTM + SENet on expression and head-pose descriptors; 3-fold CV by participant
34	Shi et al. (2022)	3	N	Y	Y	Y	Y	4 children, 19 in-home SAR sessions; XGBoost with supervised domain adaptation; session-wise chronological split
35	Silva et al. (2021)	3	N	Y	Y	ND	ND	SVM trained on 43 non-ASD participants; applied with 6 children with ASD in robot sessions
36	Alvari et al. (2021)	3	ND	Y	Y	ND	Y	OpenFace action units on home videos; statistical analysis only
37	Carpenter et al. (2020)	3	ND	Y	Y	ND	ND	Tablet computer vision (facial landmarks, expressions, attention); LASSO-selected multivariable logistic regression per movie; leave-one-out cross-validation; AUC 0.62–0.73
38	Pliska et al. (2025)	3	N	Y	Y	Y	ND	Boys only; gaze and emotion-task features; decision trees (Gini); no cross-validation or hold-out reported, accuracy computed on the study sample; ACC 81.25%, sensitivity 91.67%
39	Voss et al. (2019)	2	Y	Y	N	Y	Y	Superpower Glass randomized trial (71 children with ASD, home use); the embedded emotion classifier (8 emotions, architecture not identifiable in the main text) has an empirical accuracy of 72%; runs on the paired smartphone at about 15–20 frames per second, emotion recognized within 100 ms
40	Fan et al. (2025)	3	N	ND	Y	N	ND	EmoLand; CNN trained on CAFE + FER-2013; AI feedback accuracy 97.6% by specialist review
41	Wall et al. (2025)	3	ND	Y	Y	N	N	Pre-post TYLES app; eye-tracking of CAFE faces + ERP
45	Carter Leno et al. (2020)	4	Y	Y	Y	ND	ND	Eye-tracking subset n=46 of 189 adolescents; Dev Psychopathol 33(4) 2021, online 2020
48	Kalantarian et al. (2019)	3	N	ND	Y	ND	ND	Guess What? game videos of 8 boys with ASD (single session); frames labelled with commercial emotion classifiers combined in an ensemble (7 categories); correctly labelled frames: disgust 94%, neutral 81%, surprise 92%, scared 56%
49	Han et al. (2022)	3	ND	Y	Y	ND	ND	EEG + eye-tracking fusion with multimodal stacked denoising autoencoders (MMSDAE); 10-fold subject-independent stratified cross-validation; ACC 95.56%, sensitivity 92.5%, specificity 98%; ROC/AUC reported
51	Bochet et al. (2021)	3	Y	Y	Y	ND	Y	Eye-tracking of emotional vs neutral faces, 24 ASD vs 22 TD preschoolers

Section F. Retracted or withdrawn records

Eight records were retracted or withdrawn by their publishers. Four were identified during the V3.5 integrity check and four during the V3.6 re-screening. None participates in any denominator, and none of their results is used in the manuscript.

Record	DOI	Title	Year	Status
SCO-0013	10.1371/journal.pone.0338701	Can artificial intelligence and face recognition using deep learning detect emotions in children with autism?	2025	Retracted article (V3.5)
SCO-0026	10.1016/j.rineng.2025.106989	ERAnet: Emotion Recognition based Assistive learning network for autistic children	2025	Article removed at the request of the Executive Editors (V3.6 re-assessment)
SCO-0036	10.1038/s41598–025–97708–7	A multi-filter deep transfer learning framework for image-based autism spectrum disorder detection	2025	Retracted 8 Dec 2025 (undocumented image provenance) (V3.6 re-assessment)
SCO-0037	10.1038/s41598–025–96014–6	Leveraging artificial intelligence for diagnosis of children autism through facial expressions	2025	Retracted article (V3.5)
SCO-0190	10.1038/s41598–024–77549–6	Ambiguous facial expression detection for Autism Screening using enhanced YOLOv7-tiny model	2024	Retracted article (V3.5)
SCO-0244	10.1049/htl2.12073	Autism spectrum disorder detection using facial images: A performance comparison of pretrained convolutional neural networks	2024	Retracted article (V3.5)
SCO-0271	10.1007/s00500–023–09477-y	Real-time facial emotion recognition model based on kernel autoencoder and convolutional neural network for autism children	2024	Retracted 16 Nov 2025 (V3.6 re-assessment)
WOS-0252	10.1155/2022/2247675	RETRACTED: An AI-Enabled Internet of Things Based Autism Care System for Improving Cognitive Ability of Children with Autism Spectrum Disorders (Retracted Article)	2022	Retracted 14 Sep 2023 (V3.6, omitted 479)

Several of these retractions concern facial images of minors from non-curated web collections without a verified ASD diagnosis or documented consent. Five eligible studies use Kaggle ASD image collections (Shaik et al., Alam et al., Aljbori et al., Vedantham et al., Hosney et al.) and one uses web videos (Kurian and Tripathi). They remain in the corpus because they are not retracted, but their results carry this risk of bias and are interpreted accordingly.

Section G. Implementation and validation evidence of the added studies

The six nominal categories of the manuscript were applied to the 17 added studies with the same explicit assignment rule: longitudinal clinical validation if demonstrated; otherwise evaluation with end users; otherwise measurement on device; otherwise prototype without deployment metrics; otherwise offline evaluation; otherwise no implementation. The rule resolves membership and does not express an order of maturity.

Category	Added studies
No implementation	Alvari et al. (2021), Carter Leno et al. (2020), Bochet et al. (2021)
Offline evaluation	Kiejdo et al. (2025), Wang et al. (2025), Kurian and Tripathi (2024), Manelis-Baram et al. (2025), Kalantarian et al. (2019), Shi et al. (2022), Chen et al. (2024), Han et al. (2022)
Prototype without deployment metrics	Carpenter et al. (2020), Pliska et al. (2025)
Measurement on device	None as assigned category
Evaluation with end users	Silva et al. (2021), Voss et al. (2019), Fan et al. (2025), Wall et al. (2025)
Longitudinal clinical validation	None

Silva et al. (2021) and Voss et al. (2019) also measure frame rate or latency on the deployment device, which is evidence of measurement on device; the assignment rule places them in evaluation with end users. Voss et al. (2019) is the closest to longitudinal clinical validation: it is a randomised clinical evaluation of a deployed system, but with two time points and no longitudinal series. Of the V3.5 studies, Reed et al. (2021, no implementation) and Almeida et al. (2019, evaluation with end users) left the corpus by CI(1), and Sholikah et al. (2024, prototype without deployment metrics), Rudovic et al. (2018), Rudovic et al. (2019), Dapogny et al. (2019) and Sariyanidi et al. (2025), the last four with offline evaluation, left it by CI(3). Resulting distribution for the 31 eligible studies: no implementation, 8; offline evaluation, 14; prototype without deployment metrics, 4; measurement on device, 0; evaluation with end users, 5; longitudinal clinical validation, 0.